Crouzeix: Folie à Deux

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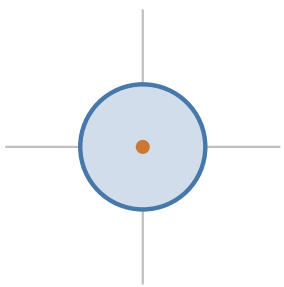
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Numerical range

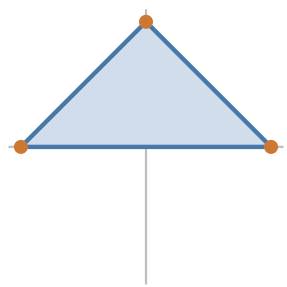
Def. Let $\mathcal{H}$ be a complex Hilbert space and $T \in \mathcal{B}(\mathcal{H})$ a bounded operator. The numerical range of T is

$$W(T) = \{\langle Tx, x\rangle : \ x \in \mathcal{H}, \ \|x\| = 1\}.$$

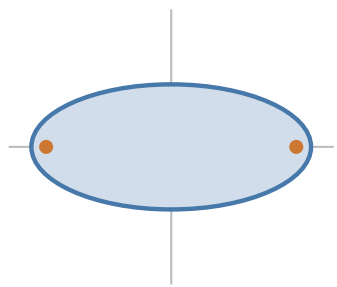
In finite dimension closed and convex (Toeplitz–Hausdorff). In infinite dimension convex, $\overline{W(T)} \supseteq \operatorname{conv} \sigma(T)$.



$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$



$\operatorname{diag}(1, i, -1)$



$\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

Orange: $\sigma(T)$. Blue: $W(T)$.

Spectral sets

Def. A set $S \subseteq \mathbb{C}$ is called a K-spectral set for an operator $T \in \mathcal{B}(\mathcal{H})$ if

$$\|p(T)\| \leq K \sup_{z \in S} |p(z)| \qquad \forall\, p \in \mathbb{C}[z].$$

(S bounded, polynomially convex.)

von Neumann's inequality. The disc is a spectral set for any contraction:

$$\|p(T)\| \leq \sup_{z \in \mathbb{D}} |p(z)|, \qquad \|T\| \leq 1.$$

(Also the closed half plane if $\operatorname{Re} T = \frac{T+T^*}{2} \geq 0$.)

von Neumann	Sz.-Nagy
function theory	dilation theory
Blaschke products	unitary dilation of T

Crouzeix conjecture

Conjecture (Crouzeix 2004)

Let T be a bounded operator on a Hilbert space $\mathcal{H}$. The numerical range $\overline{W(T)}$ is a 2-spectral set for T.

Remark. This conjectured bound is sharp:

$$W\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = D(0, \tfrac{1}{2}), \qquad p(z) = z,$$

$$\left\| \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right\| = 1 = 2 \cdot \tfrac{1}{2} = 2 \sup_{D(0,1/2)} |p(z)|.$$

Antecedents.

- von Neumann, only for contractions.
- Okubo–Ando 1975, Berger–Stampfli 1967: the disc is a 2-spectral set.

Constants

Delyon–Delyon	**1999**	$C_\Omega < \infty$	
Crouzeix	**2004**	2	**only for** 2×2
Crouzeix	**2007**	11.08	
Crouzeix–Palencia	**2017**	$1 + \sqrt{2}$	**completely bounded**
Jin	**July 2026**	2	
Lorist–Schwenninger	**Aug 2026**	2	

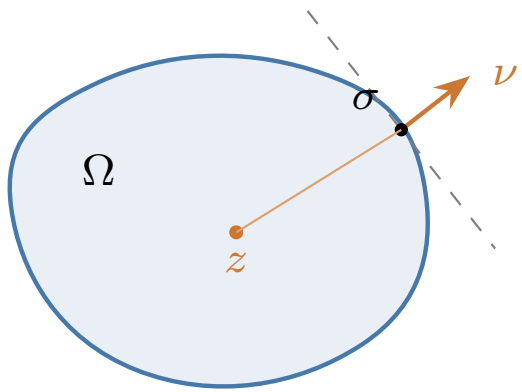
Technique. The double layer potential.

Double layer potential

Let $\Omega \subset \mathbb{C}$ be an open bounded convex set with smooth boundary $\partial\Omega$ and $\overline{W(T)} \subset \Omega$. $A(\Omega)$: holomorphic on Ω, continuous on $\overline{\Omega}$, sup norm. $C(\partial\Omega)$: continuous on $\partial\Omega$, sup norm. For $\sigma \in \partial\Omega$ denote the outward pointing normal by $\nu = \nu(\sigma)$, and if $z \in \Omega$ define the double layer potential by

$$\mu(\sigma, z) = \frac{1}{2\pi}\Big(\underbrace{\frac{\nu}{\sigma - z}}_{\text{first layer}} + \underbrace{\frac{\bar{\nu}}{\bar{\sigma} - \bar{z}}}_{\text{second layer}} \Big)$$

$= \frac{1}{\pi} \operatorname{Re} \frac{\nu}{\sigma - z}$. First layer: the Cauchy kernel.



- $\mu\, ds = \frac{1}{\pi}\, d \arg(\sigma - z)$
- $\mu \geq 0$
- $\int_{\partial\Omega} \mu\, ds = 2$

Why use? The Cauchy layer leads to a bound on $f(z)$. If we can control the conjugate we win. (Adjoint when applying functional calculus.)

Double layer potential, cont.

Let

$$C(f,T) = \frac{1}{2\pi i}\int_{\partial\Omega} f(\sigma)(\sigma - T)^{-1}\,d\sigma \in \mathcal{B}(\mathcal{H})$$

and

$$S(f,T) = C(f,T) + C(\bar{f},T)^* \in \mathcal{B}(\mathcal{H}).$$

By definition S is the function calculus integration against the double layer potential $\mu(\sigma, z)$, i.e.

$$S(f,T) = \int_{\partial\Omega} f(\sigma)\,\mu(\sigma,T)\,ds.$$

Lemma (Crouzeix–Palencia 2017)

For $f \in C(\partial\Omega)$, $\|S(f,T)\| = \|f(T) + C(\bar{f},T)^*\| \leq 2\|f\|_{\partial\Omega}$.

So the question we must answer is: how does $\|S(f,T)\|$ control $\|f(T)\|$? In particular, can we estimate $C(\bar{f},T)^*$ or make it vanish all together?

Positivity and the double layer formula

Consider

$$2\Phi(f) := f(T) + C(\bar{f}, T)^* = f(T) + \alpha(f)(T)^*,$$

where

$$\alpha(f)(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{f(\sigma)}}{\sigma - z}\, d\sigma.$$

Note that $\alpha : A(\Omega) \to A(\Omega)$ is antilinear and $\alpha(1) = 1$.

Positivity of Φ says $2\|\Phi(f)\| \leq 2\|f\|_{\partial\Omega}$: the result from the previous slide, found in Crouzeix–Palencia.

Scalar case: $2\Phi(f)(z) = f(z) + \overline{\alpha(f)(z)}$. Disc: $\alpha(f) = \overline{f(0)}$.

An alternate approach (not taken by either ChatGPT proof) is to "hack" the formula by studying extremal functions: f extremal, $\|f(T)\| \leq \|2\Phi(f)\|$, which proves Crouzeix. (Bickel, Gorkin, Greenbaum, Overton.)

Crouzeix–Palencia, $1+\sqrt{2}$

$K = C_\Omega$, $\|f\| \le 1$, $T = f(A)$, $g = \alpha(f)$, $\|g\| \le \|f\|$ (Ω convex).

$$TT^*TT^* = T\left(T + g(A)^*\right)^* TT^* - (fgf)(A)\,T^*$$

$$K^4 \le 2K^3 + K^2, \qquad K \le 1+\sqrt{2}.$$

(Ransford–Schwenninger)

Uses only $\|f(A) + g(A)^*\| \le 2$ and $\|g\| \le \|f\|$. Sharp under these (Ransford–Schwenninger 2018):

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \Omega = D(0, \tfrac{1}{4}) \cup D(1, \tfrac{1}{4}),$$

$g = -f(0)$, $-f(1)$ on the two discs; $h = \mp 1$: $\|h(T)\| = 1+\sqrt{2}$.

Here $g(1) = -1$. The Cauchy transform: $\alpha(1) = 1$.

Equivalent formulation

Theorem (Twisted real part)

Let $\mathcal{A}$ be a unital commutative operator algebra, let $\theta : \mathcal{A} \to \mathcal{B}(\mathcal{H})$ be a bounded homomorphism and let $\alpha : \mathcal{A} \to \mathcal{A}$ be a unital antilinear map such that the unital linear map

$$\theta_\alpha : \mathcal{A} \to \mathcal{B}(\mathcal{H}), \qquad a \mapsto \tfrac{1}{2}(\theta(a) + \theta(\alpha(a))^*)$$

is contractive. Then $\|\theta\| \leq 2$.

- Ransford–Schwenninger (2018) show that the twisted real part theorem $\Rightarrow$ Crouzeix.
- Michael Hartz (2026) proves the theorem using Jin's ChatGPT solution.

Un

Function theory

Function theory

Jin's GitHub; unpublished notes by Hartz.

Theorem (Holomorphic $\to$ resolvent transfer)

Let $T \in \mathcal{B}(\mathcal{H})$ with $\sigma(T) \subset \overline{\mathbb{D}}$. Let $H : \mathbb{D} \to \mathcal{B}(\mathcal{H})$ be a holomorphic function such that $H(0) = I$ and $\operatorname{Re} H(z) \geq 0$ for all $z \in \mathbb{D}$. Assume that

$$H(z) - (I - zT)^{-1}$$

commutes with T^* for all $z \in \mathbb{D}$. Then $\|T\| \leq 2$.

- Hartz shows that holomorphic $\to$ resolvent transfer $\Rightarrow$ twisted real part.
- $H(z) - (I - zT)^{-1} = 0$: Okubo–Ando.

Proof

$\Delta(z) := H(z) - (I - zT)^{-1}$, $R(z) := (I - zT)^{-1}$. On $H^2 \otimes \mathcal{H}$: $\operatorname{Re} M_H \geq 0$, $M_H = M_R + M_\Delta$.

$N_W := \{F = \sum F_n z^n : \sum W^n F_n = 0\}$, $\sigma(W) \subset \mathbb{D}$.

- $\Delta(z)W = W\Delta(z) \Rightarrow M_\Delta N_W \subset N_W$.
- $\Delta(0) = 0$, Γ the kernel of N_W at $0 \Rightarrow \Gamma^* M_\Delta \Gamma = 0$.

$W = \frac{1}{2}T^*$:

$$\Gamma^*(M_R + M_R^*)\Gamma = \Gamma^*(M_H + M_H^*)\Gamma \geq 0.$$

$P = \sum 4^{-n} T^{*n} T^n$, $Q = \sum 2^{-n} T^{*n} T^n$, $Y = Q - P \geq \frac{1}{4} T^* T$:

$$4Y - YP - PY \geq 0 \;\Rightarrow\; P \leq 2I \;\Rightarrow\; T^*T \leq T^*PT = 4(P - I) \leq 4I.$$

Jin: sample the Herglotz kernel of H at $\lambda_i/2$ and at 0.

Deux

Dilation theory

Dilation theory

Lemma (Lorist–Schwenninger, ChatGPT)

Let $\mathcal{H}$ be a Hilbert space and $T \in \mathcal{B}(\mathcal{H})$. Suppose there exists a Hilbert space $\mathcal{K}$, a contraction $Q \in \mathcal{B}(\mathcal{K})$ and an isometry $V \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that the operators

$$E_n := 2V^* Q^{*n} V - T^{*n}, \qquad n \in \mathbb{N},$$

are uniformly bounded and commute with T. Then $\|T\| \leq 2$.

- $E_n = 0$: 2-dilation of T^*, $\|T\| \leq 2$.
- $\mathcal{K} = L^2(\partial\Omega, ds; \mathcal{H})$, $Vx = \frac{1}{\sqrt{2}} \mu(\cdot, T)^{1/2} x$, $Q = M_f$, $\|f\|_\infty = 1$, $T = f(A)$:

  $$V^* Q^n V = \Phi(f^n), \qquad E_n = 2\Phi(f^n)^* - f(A)^{*n} = \alpha(f^n)(A).$$

- E_n commutes with $f(A)$; $\|E_n\| \leq \|\theta\| \|\alpha\|$.

Proof

$\kappa := \|T\| > 1$, $T^*Tx = \kappa^2 x$, $\|x\| = 1$ (ultrapower).

$$m_n := \mathrm{Re}\langle E_n T^n x, x\rangle, \qquad u := Q^*VTx - \kappa Vx.$$

$$\|u\|^2 \le 2\kappa^2 - \kappa^3 - \kappa m_1 \qquad (n = 1)$$

$$\kappa m_1 \ge -\frac{\|u\|^2}{(\kappa-1)^2} \qquad (\text{telescope } \kappa m_n - m_{n+1},\ \text{all } n)$$

$$\|u\|^2\big(1 - (\kappa-1)^{-2}\big) \le \kappa^2(2-\kappa).$$

First line alone: $\kappa \le 1 + \sqrt{1 - m_1}$.
$\|\alpha\| \le 1 \Rightarrow m_1 \ge -1 \Rightarrow 1 + \sqrt{2}$. $m_1 < 0$ for some 3×3.

Un and Deux

	Un	Deux
positivity	$\operatorname{Re} H \geq 0$	$\Phi = V^* \pi(\cdot) V$
all powers	$H(w) = I + 2\sum_m w^m \Phi(f^m)$	E_n, all n
companion	$\sum_m w^m \alpha(f^m)(T)^*$	$\alpha(f^n)(T)$
used	commutes with T^*	commutes with T
removal	$\Gamma^* M_\Delta \Gamma = 0$	telescoping

Crouzeix–Palencia: $\|\alpha(f)\| \leq \|f\|$. Un, Deux: $\alpha(f^n)(T) \in \{g(T)\}$.

The canary

- Jin: ChatGPT proposed sampling at $\lambda_i/2$ and at 0.
- Lorist–Schwenninger: GPT-5.6, 25–26 July.
  $\Phi(f^{n+1})^* f(A)x - \|f(A)\|\Phi(f^n)^* x$ controlled uniformly in n.
  Saw Jin 28 July.
- Hartz: ChatGPT to discuss Jin.

Same step twice: one $f \to$ all f^n.

Completely bounded

Conjecture

For every $m \geq 1$ and $F = [f_{ij}] \in M_m(A(\Omega))$,
$\|F(T)\| \leq 2 \sup_{z \in \Omega} \|F(z)\|$.

- Known: $1 + \sqrt{2}$ (Crouzeix–Palencia); 2 for discs, weighted shifts (Crouzeix–Greenbaum 2025).
- $M_m(A(\Omega))$ not commutative. Un, Deux need the companion to commute with $f(T)$.
- Crouzeix–Palencia: only $\|\alpha\| \leq 1$.
- Last week: [reference]. OpenAI SlopDrop (Burnstein, Iosevich, Krause, Yavicoli).

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Appendix: Γ

$K_W := E_W^*$, $(K_W x)(z) = (I - zW^*)^{-1}x$, $K_0 x \equiv x$. $W = \frac{1}{2}T^*$,
$P = K_W^* K_W \geq I$, $K_W P^{-1} K_W^* =$ projection onto $\operatorname{ran} K_W = N_W^\perp$.

$$\Gamma = (I - K_W P^{-1} K_W^*) K_0 = K_0 - K_W P^{-1} \qquad (K_W^* K_0 = I).$$

$(I - zT)^{-1}(I - zT/2)^{-1} = 2(I - zT)^{-1} - (I - zT/2)^{-1}$:

$$M_R K_W = 2K_{T^*} - K_W, \qquad M_R K_0 = K_{T^*}, \qquad M_R^* K_0 = K_0.$$

$Q = K_W^* K_{T^*}$:

$$\begin{pmatrix} K_W^* \\ K_0^* \end{pmatrix} (M_R + M_R^*) \begin{pmatrix} K_W & K_0 \end{pmatrix} = \begin{pmatrix} 4Q - 2P & Q + I \\ Q + I & 2I \end{pmatrix},$$

$$\Gamma^*(M_R + M_R^*)\Gamma = P^{-1}(4Y - YP - PY)P^{-1}.$$

Appendix: $4Y - YP - PY \geq 0 \Rightarrow \|T\| \leq 2$

$P = \sum_{n \geq 0} 4^{-n} T^{*n} T^n \geq I$, $Y \geq \frac{1}{4} T^* T \geq 0$.

$Px = \alpha x$, $\|x\| = 1$, $\alpha > 2$:

$$0 \leq \langle (4Y - YP - PY)x, x \rangle = (4 - 2\alpha)\langle Yx, x \rangle,$$

so $Yx = 0$, so $Tx = 0$, so $Px = x$. (Approximate eigenvectors, or ultrapower.) So $I \leq P \leq 2I$.

$P = I + \frac{1}{4} T^* P T$:

$$T^* T \leq T^* P T = 4(P - I) \leq 4I.$$

Appendix: the two estimates

$E_nT = TE_n$, $T^*Tx = \kappa^2 x$:

$$\begin{aligned}
\kappa m_n - m_{n+1} &= \operatorname{Re}\langle T^n(\kappa E_n - E_{n+1}T)x, x\rangle \\
&= \kappa(\kappa-1)\|T^{*n}x\|^2 - 2\operatorname{Re}\langle V^*Q^{*n}u, T^{*n}x\rangle \\
&= \kappa(\kappa-1)\Big\|T^{*n}x - \frac{V^*Q^{*n}u}{\kappa(\kappa-1)}\Big\|^2 - \frac{\|V^*Q^{*n}u\|^2}{\kappa(\kappa-1)} \\
&\geq -\frac{\|u\|^2}{\kappa(\kappa-1)}.
\end{aligned}$$

Multiply by κ^{-n+1}, sum over $n \geq 1$; (m_n) bounded, so the sum telescopes to κm_1.

$\|u\|^2 = \|Q^*VTx\|^2 + \kappa^2 - 2\kappa\operatorname{Re}\langle V^*Q^*VTx, x\rangle$, $\|Q^*VTx\| \leq \kappa$, $2V^*Q^*V = T^* + E_1$:

$$\|u\|^2 \leq 2\kappa^2 - \kappa\operatorname{Re}\langle T^*Tx, x\rangle - \kappa m_1 = 2\kappa^2 - \kappa^3 - \kappa m_1.$$